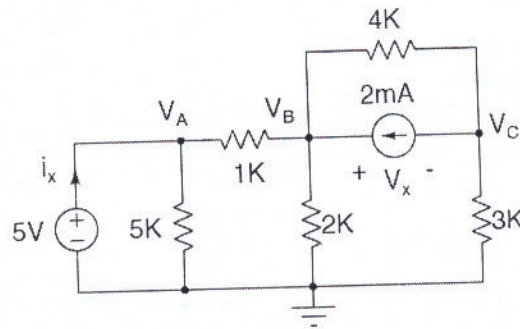


EEE-211 - FALL 2019-2020
MIDTERM EXAM

Question 1 [20 Points]:



- (a) For the circuit above, find V_A , V_B , V_C using nodal analysis.
(b) Using your result in part (a), find the current i_x and the voltage V_x marked in the circuit.

a) $V_A = 5V$

KCL @ B : $\frac{V_B - 5}{1K} + \frac{V_B}{2K} + \frac{V_B - V_C}{4K} - 2mA = 0$
(4) (2) (4K)

$7V_B - V_C = 28 \Rightarrow \boxed{V_C = 7V_B - 28}$

KCL @ C : $\frac{V_C - V_B}{4K} + \frac{V_C}{3K} + 2mA = 0$
(3) (4) (12K)

$7V_C - 3V_B + 24 = 0$

$7(7V_B - 28) - 3V_B + 24 = 0$

$46V_B = 172 \Rightarrow \boxed{V_B \approx 3.74V}$

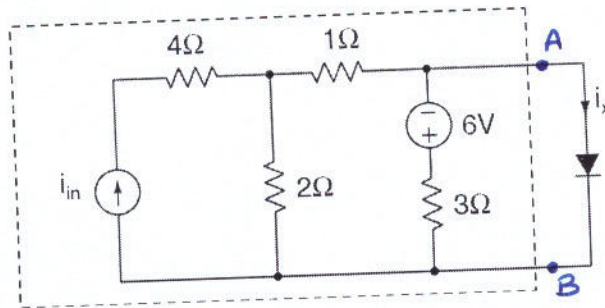
Then, $V_C = 7 \times 3.74 - 28 \Rightarrow \boxed{V_C \approx -1.83V}$

b) KCL @ A : $i_x = \frac{V_A}{5K} + \frac{V_A - V_B}{1K} = \frac{5}{5K} + \frac{5 - 3.74}{1K}$

$i_x = 2.26 \text{ mA}$

$V_x = V_B - V_C = 3.74 - (-1.83) \Rightarrow \boxed{V_x = 5.57V}$

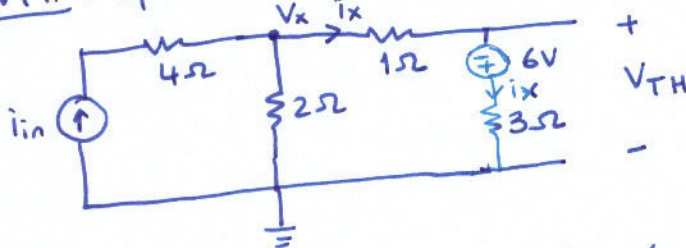
Question 2 [30 points]:



- (a) For the circuit above, find the Thevenin equivalent of the circuit inside the dashed box, as seen by the diode. Express V_{TH} in terms of i_{in} . Draw the Thevenin equivalent circuit.
- (b) Assume that the diode has a threshold voltage of $V_0 = 0.6 V$ and $i_{in}(t) = 6 \sin(2\pi 1000t) A$. Find and plot $i_x(t)$. Mark all important time points and current values in your graph.
- (c) For the same circuit, assume that $i_{in} = 1 A$ (DC) and the diode is replaced with a capacitor of $C = 2 \mu F$. The initial voltage for the capacitor is $V_C(0) = 0 V$. Find and plot $i_x(t)$ for $t > 0$. Mark all important time points and current values in your graph.

Hint: For parts (b) and (c), use your result from part (a).

a) V_{TH} : open circuit voltage between A-B



$$\text{KCL @ X: } -i_{in} + \frac{V_x}{2} + \frac{V_x + 6}{4} = 0$$

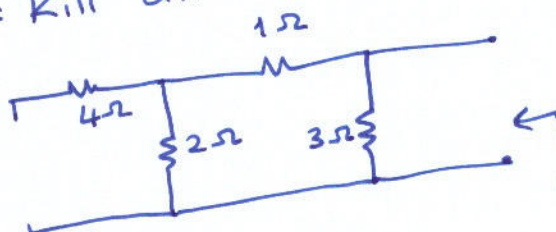
(4) (2)

$$3V_x = 4i_{in} - 6 \Rightarrow V_x = \frac{4}{3}i_{in} - 2$$

$$i_x = \frac{V_x + 6}{4} = \frac{\frac{4}{3}i_{in} - 2 + 6}{4} = \frac{i_{in}}{3} + 1$$

$$V_{TH} = 3i_x - 6 = i_{in} - 3 \Rightarrow \boxed{V_{TH}(t) = i_{in}(t) - 3V}$$

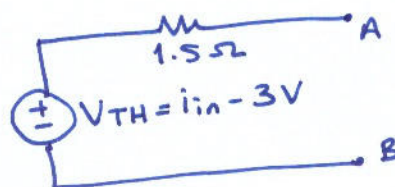
Req: Kill all sources.



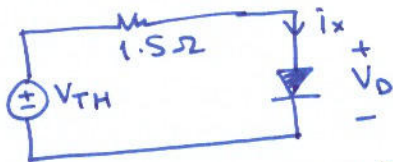
$$R_{eq} = 3 \parallel (1+2)$$

$$\boxed{R_{eq} = 1.5 \Omega}$$

Thevenin equivalent:



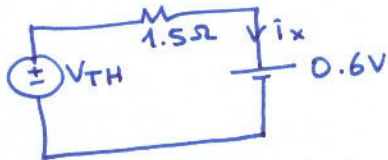
b)



$$V_{TH} = 6\sin(2\pi 1000t) - 3V \quad T = 1ms$$

* when diode is OFF, $i_x(t) = 0$

* when diode turns ON: $V_D = 0.6V$



Condition: $i_x \geq 0$

$$4\sin(2\pi 1000t) - 2.4 \geq 0$$

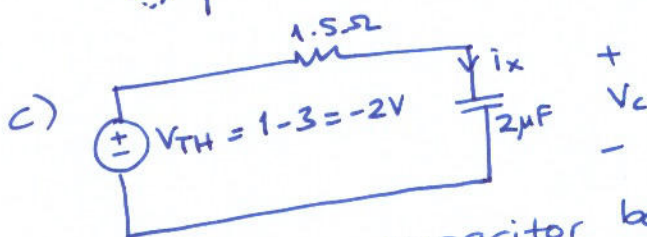
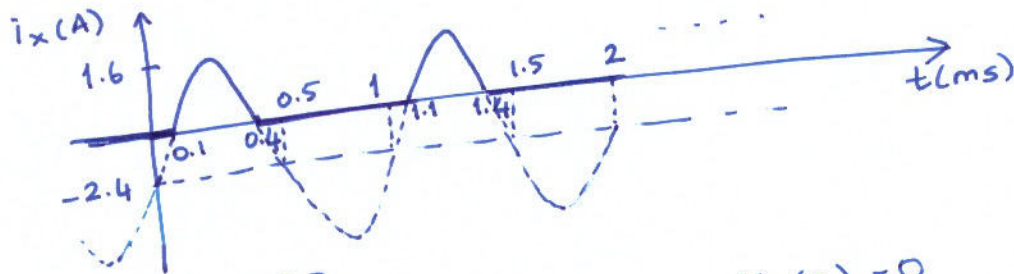
$$\sin(2\pi 1000t) \geq 0.6$$

$$\frac{a\sin(0.6)}{2\pi \cdot 1000} + nT < t < \frac{T}{2} - \frac{a\sin(0.6)}{2\pi \cdot 1000} + nT$$

$$0.1 + n < t < 0.4 + n, \quad n \in \mathbb{Z}$$

where t is in ms

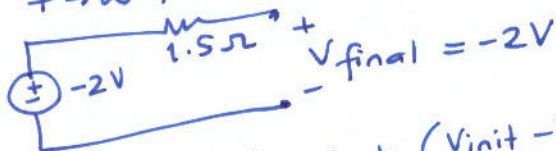
so, $i_x(t) = \begin{cases} 4\sin(2\pi 1000t) - 2.4A, & 0.1 + n < t(\text{ms}) < 0.4 + n, n \in \mathbb{Z} \\ 0, & \text{otherwise} \end{cases}$



$$V_C(0) = 0$$

$$\tau = RC = 1.5 \times 2 \times 10^{-6} = 3\mu s$$

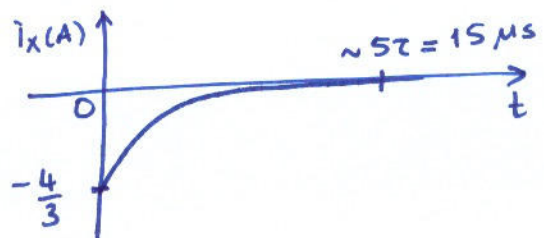
* As $t \rightarrow \infty$, capacitor behaves like open circuit



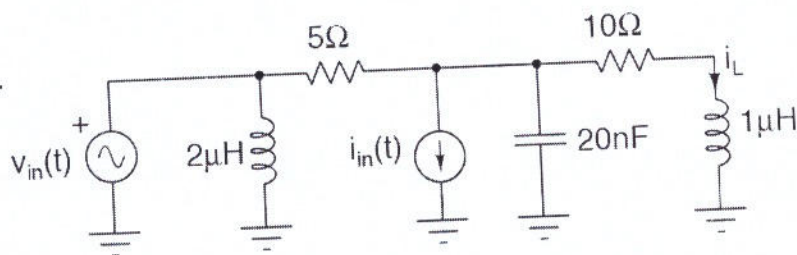
* so, $V_C(t) = V_{final} + (V_{init} - V_{final})e^{-t/\tau} = -2 + 2e^{-t/\tau} V, t \geq 0$

* Then, $i_x(t) = i_C(t) = C \frac{dV_C}{dt} = -\frac{C}{\tau} 2e^{-t/\tau} = -\frac{2 \cdot 10^{-6}}{3 \cdot 10^{-6}} \cdot 2e^{-t/\tau}$

$$i_x(t) = -\frac{4}{3}e^{-t/\tau} A, t \geq 0$$



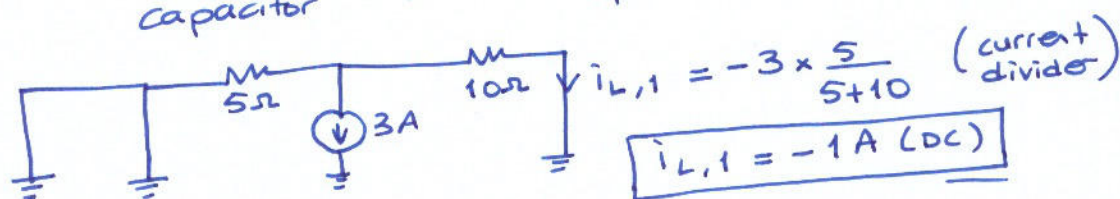
Question 3 [20 points]:



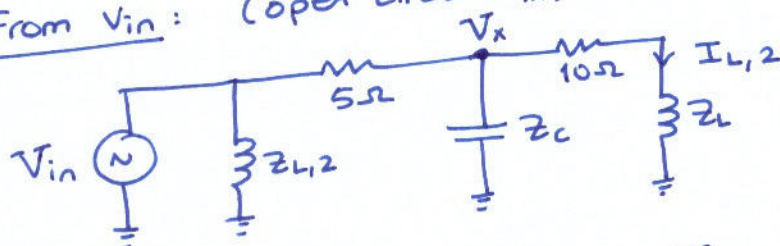
For the circuit above, $v_{in}(t) = 3 \cos(10^7 t + \pi/3) V$ and $i_{in}(t) = 3 A (DC)$, for $-\infty < t < \infty$. Find $i_L(t)$.

Use superposition:

* From i_{in} : (short circuit v_{in})
 Since i_{in} is a DC source for $-\infty < t < \infty$
 inductor acts like short circuit
 capacitor " " open "



* From v_{in} : (open circuit i_{in})



$$\omega = 10^7 \text{ rps}$$

$$Z_L = j\omega L = j10^7 \cdot 10^{-6} = j10 \Omega$$

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j10^7 \cdot 20 \cdot 10^{-9}} = -j5 \Omega$$

$$v_{in} = 3e^{j\pi/3} V$$

KCL at X: $\frac{V_x - V_{in}}{5} + \frac{V_x}{Z_C} + \frac{V_x}{10 + Z_L} = 0$

$$\frac{V_x - V_{in}}{5} + \frac{V_x}{-j5} + \frac{V_x}{10(1+j)} = 0$$

(4) (j4) (1-j)

$$V_x (4 + j4 + 1 - j) = 4V_{in} \Rightarrow V_x = \frac{4V_{in}}{5 + 3j}$$

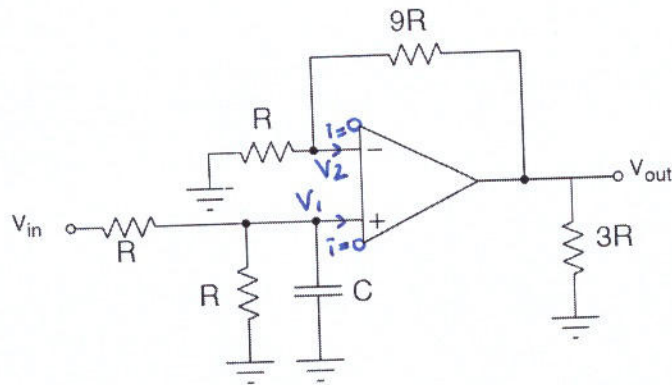
$$I_{L,2} = \frac{V_x}{10(1+j)} = \frac{4 \cdot 3 \cdot e^{j\pi/3}}{10 \cdot \sqrt{2} e^{j\pi/4} \cdot \sqrt{34} e^{j\text{atan}(3/5)}}$$

$$= \frac{3}{5\sqrt{17}} e^{j(\pi/3 - \pi/4 - \text{atan}(3/5))} \approx 0.15 e^{-j0.28}$$

$$i_{L,2}(t) = 0.15 \cos(10^7 t - 0.28) A$$

* Finally, $i_L(t) = i_{L,1} + i_{L,2}(t) = -1 + 0.15 \cos(10^7 t - 0.28) A$
 OR
 $i_L(t) = -1 + 0.15 \cos(10^7 t - 16^\circ) A$

Question 4 [30 points]:



For the circuit shown above, assume that $v_{in}(t)$ is sinusoidal and the OPAMP is not saturated.
 $R = 1 \text{ k}\Omega$ and $C = 1 \text{ nF}$.

- (a) Find the transfer function $H(\omega) = \frac{V_{out}}{V_{in}}$ in terms of ω , R , C (i.e., do not insert component values in your result yet). Express the transfer function in one of the following forms:

$$A \frac{1}{1+jX} \quad , \quad A \frac{jX}{1+jX} \quad , \quad A \frac{1+jX}{1+jY}$$

- (b) What kind of filter is this? What is the maximum gain of this circuit in dB? What is the cut-off frequency in units of Hz?
- (c) Plot $|H(\omega)|$ in logarithmic scale. Clearly show low-frequency and high-frequency asymptotes. Mark the 3-dB corner frequency and the magnitude of the transfer function at that frequency.
- (d) Find the frequency (in Hz) at which the magnitude of $v_{out}(t)$ is 20 times lower than the magnitude of $v_{in}(t)$. Make simplifying approximations when needed. **Hint:** Use the plot from part (c).

a) Since OPAMP is not saturated, $V_1 = V_2$

* $V_1 = \frac{R \parallel \frac{1}{j\omega C}}{R + (R \parallel \frac{1}{j\omega C})} \cdot V_{in}$ (voltage divider)

$$= \frac{(\frac{1}{R} + j\omega C)^{-1}}{R + (\frac{1}{R} + j\omega C)^{-1}} V_{in} = \frac{\frac{R}{1+j\omega RC}}{R + \frac{R}{1+j\omega RC}} \cdot V_{in} = \frac{1}{2+j\omega RC} \cdot V_{in}$$

* $V_2 = \frac{R}{R+9R} \cdot V_{out} = \frac{1}{10} V_{out}$

* $V_1 = V_2 \Rightarrow \frac{1}{2+j\omega RC} V_{in} = \frac{1}{10} V_{out}$

$$H(\omega) = \frac{V_{out}}{V_{in}} = \frac{10}{2+j\omega RC} = \boxed{5 \cdot \frac{1}{1+j\omega \frac{RC}{2}}}$$

b) $|H(\omega)| = \frac{5}{\sqrt{1+(\frac{\omega RC}{2})^2}}$ $\Leftarrow$ This is a low-pass filter.

@ $\omega=0$, $|H(\omega)| = 5$, gain is maximum.
 $20 \cdot \log_{10} 5 \approx \boxed{14 \text{ dB}}$

$$@ \omega_c, |H(\omega_c)| = \frac{\max |H(\omega)|}{\sqrt{2}} = \frac{5}{\sqrt{2}}$$

$$\frac{5}{\sqrt{2}} = \frac{5}{\sqrt{1 + (\frac{\omega_c RC}{2})^2}} \Rightarrow \frac{\omega_c RC}{2} = 1 \Rightarrow \omega_c = \frac{2}{RC}$$

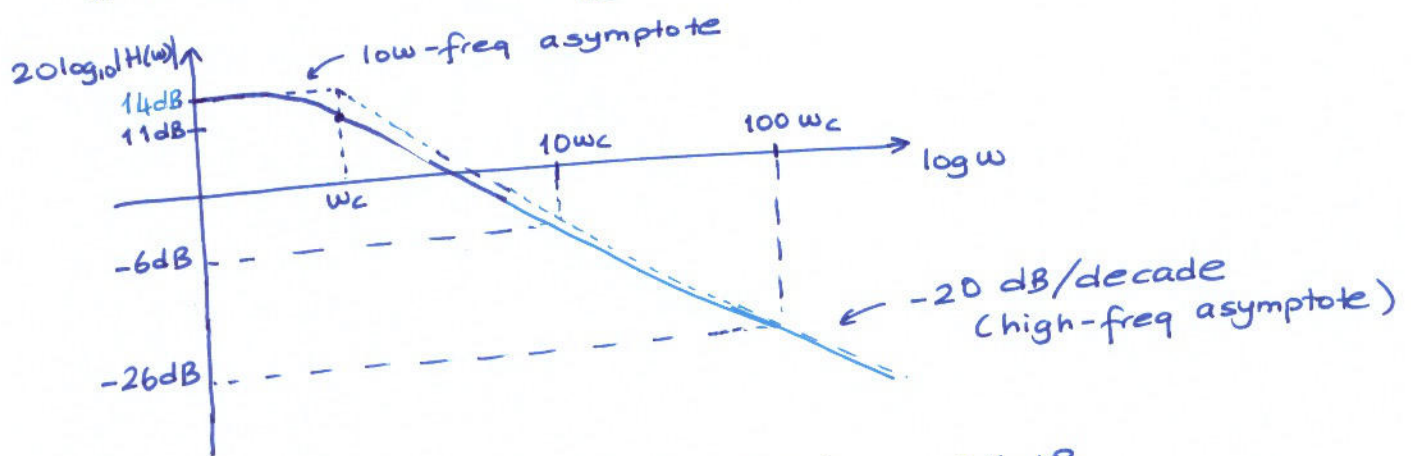
$$f_c = \frac{\omega_c}{2\pi} = \frac{2}{2\pi RC} = \frac{1}{\pi 10^3 \cdot 10^{-9}} = \boxed{318.3 \text{ kHz}}$$

$$c) |H(\omega)| = \frac{5}{\sqrt{1 + (\frac{\omega}{\omega_c})^2}}, \text{ where } \omega_c = \frac{2}{RC}$$

$$* \text{ when } \omega \ll \omega_c, |H(\omega)| \approx 5 \Rightarrow 14 \text{ dB}$$

$$* \text{ when } \omega \gg \omega_c, |H(\omega)| \approx 5 \frac{\omega_c}{\omega} \Rightarrow -20 \text{ dB/decade slope}$$

$$* \text{ at } \omega = \omega_c, |H(\omega)| = \frac{5}{\sqrt{2}} \Rightarrow 14 - 3 = 11 \text{ dB}$$



$$d) |H(\omega_x)| = \frac{1}{20} \Rightarrow 20 \log_{10} \left(\frac{1}{20} \right) = -26 \text{ dB}$$

Based on the plot above, this happens at $\sim 100 \omega_c$.
So, $\omega_x \approx 100 \omega_c$

$$f_x \approx 100 f_c = \boxed{31.8 \text{ MHz}}$$

Alternatively, $|H(\omega_x)| = \frac{1}{20} = \frac{5}{\sqrt{1 + (\frac{\omega}{\omega_c})^2}}$

$$1 + \left(\frac{\omega_x}{\omega_c} \right)^2 = 10000$$

$$\omega_x = \sqrt{9999} \omega_c$$

$$f_x = \sqrt{9999} f_c \approx \boxed{31.8 \text{ MHz}}$$